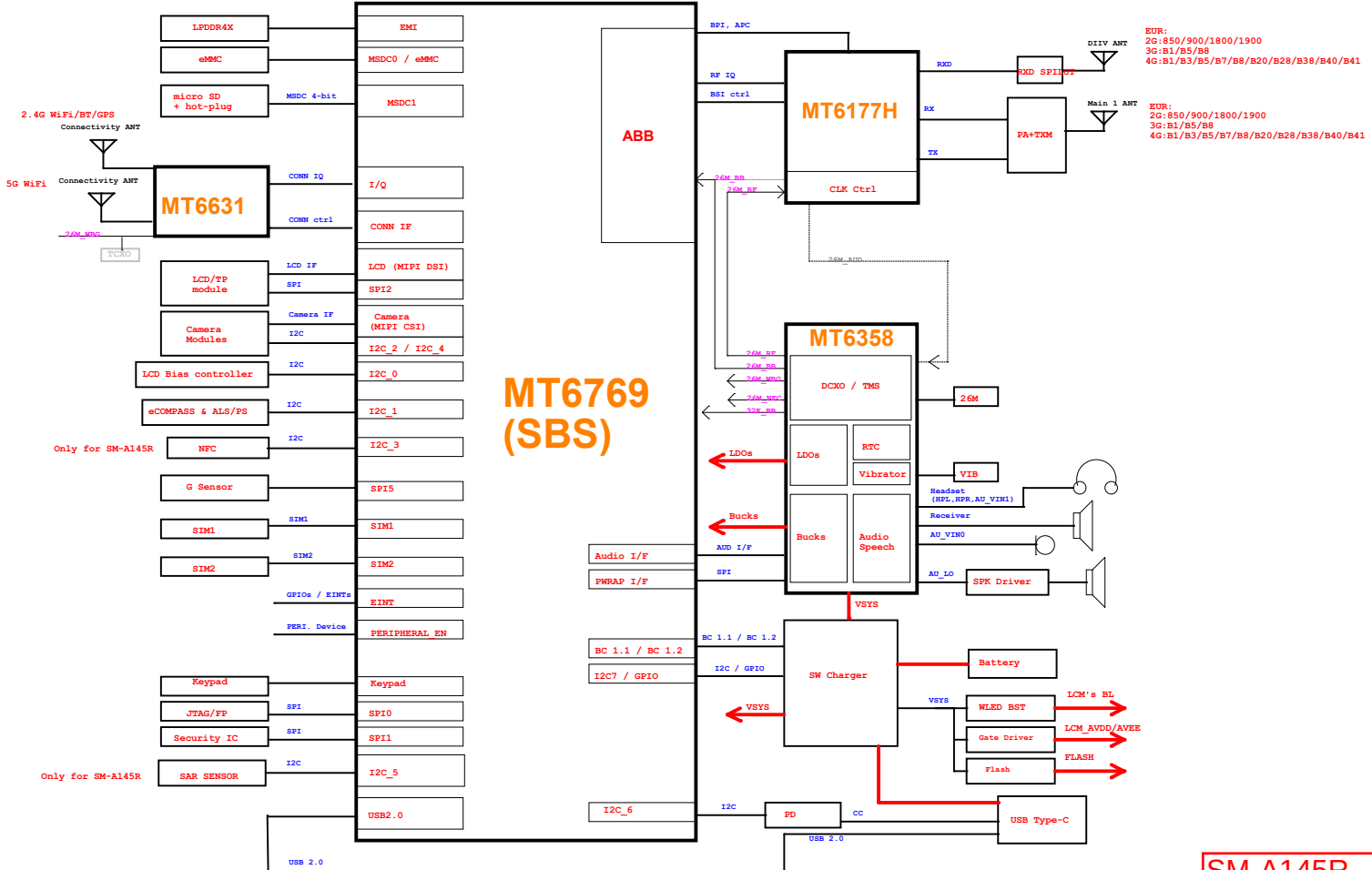


SIM-A145R

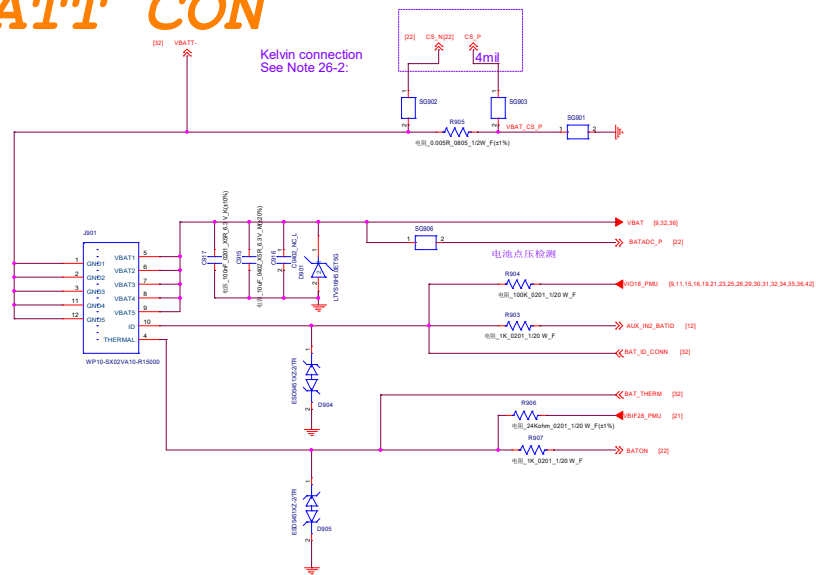


## I2C Configuration Table

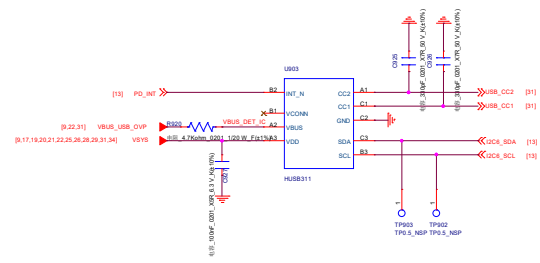
| Function              | I2C Slave Address (7-bit mode) |
|-----------------------|--------------------------------|
| Page 15 : Charger IC  |                                |
| Page 16 : LCM BIAS IC |                                |
| Page 21 : SMART PA    |                                |
|                       |                                |
|                       |                                |
| Page 23 : Camera      |                                |
|                       |                                |
|                       |                                |
| E-COMPASS             |                                |
| ALS&PS SENSOR         |                                |
| NFC                   |                                |
|                       |                                |

- 01.Block Diagram
- 02.I2C\_ID\_Overview
- 03.Index
- 04.BB\_POWER\_PDN
- 05.BB\_POWER\_IO
- 06.BB\_1\_GPIO
- 07.BB\_2\_GPIO
- 08.BB\_3\_EMI
- 09.BB\_AUXADC
- 10.POWER\_MT6357\_Buck
- 11.POWER\_MT6357\_LDO
- 12.POWER\_MT6357\_IF
- 13.POWER\_MT6357\_Audio
- 14.POWER\_BAT\_CON
- 15.CHARGE\_IC+CC
- 16.LCD BACKLIGHT+LCD BIAS
- 17.FLASHLED DRIVER
- 18.Memory\_eMMC\_LPDDR3
- 19.POWER\_ThirdParty\_Powers
- 20.Audio\_EARJACK
- 21.Audio\_MIC\_Receiver
- 22.Audio\_K8\_PA
- 23.LCM\_FPS
- 24.CAMERA
- 25.LINK
- 26.SIM\_SD
- 27.SideKey\_Flash\_LED
- 28.Sensor
- 29.Test Point\_JTAG
- 30.RF\_MT6177\_Pin\_Out
- 31.RF\_MT6177\_RF\_TX
- 32.RF\_MT6177\_RF\_PRX
- 33.RF\_MT6177\_RF\_DRX
- 34.RF\_ANT
- 35.WCN\_FEM
- 36.CONNECTIVITY\_CONSYS\_MT6631
- 37.SAR\_Sensor

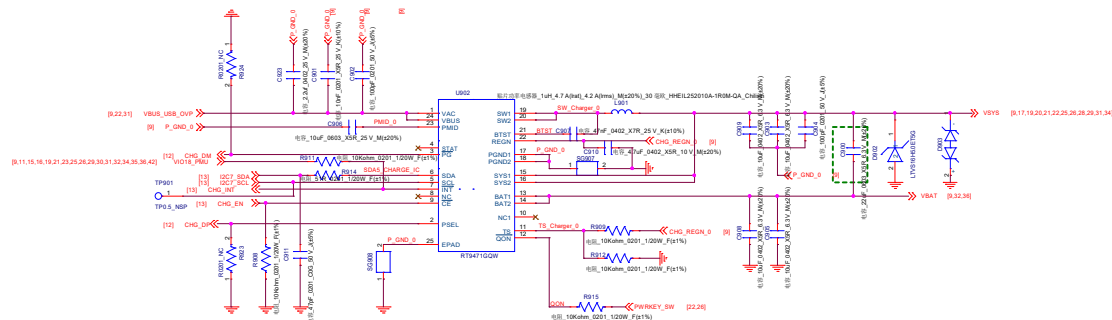
# BATT CON



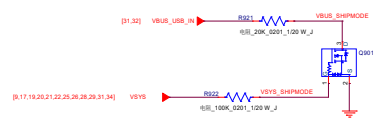
# PD

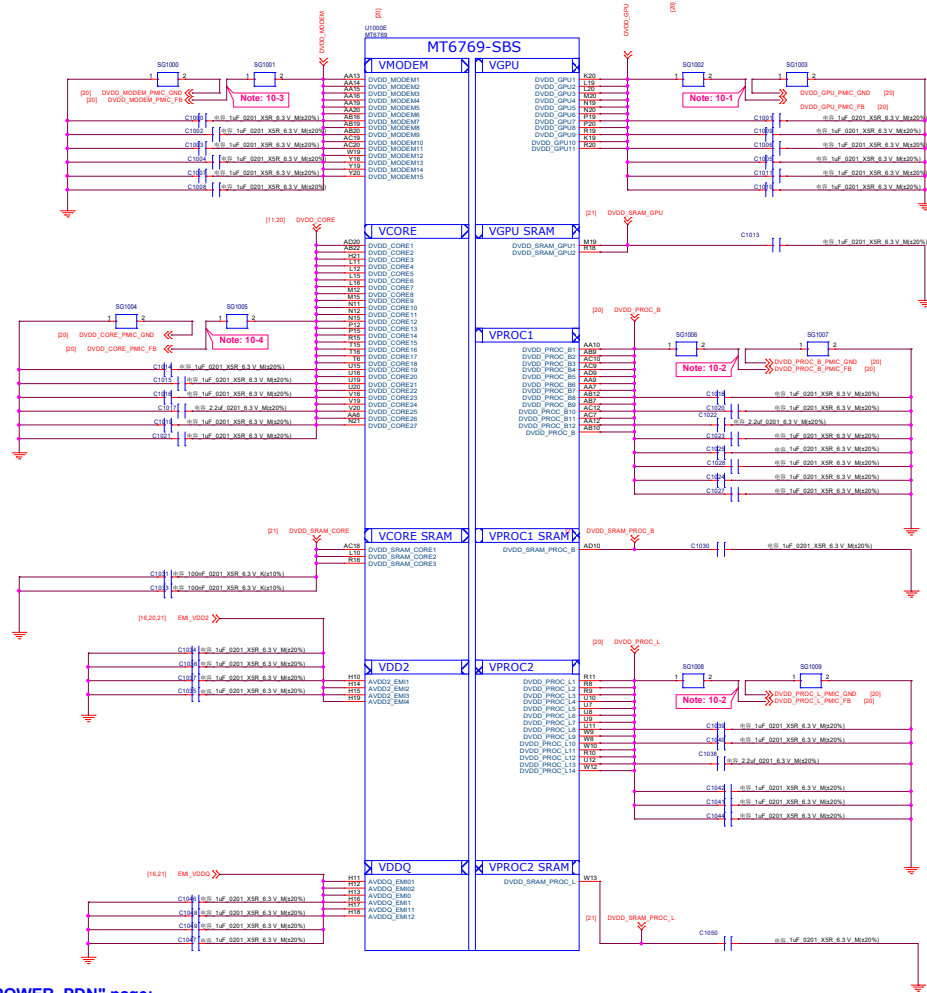


# Charger



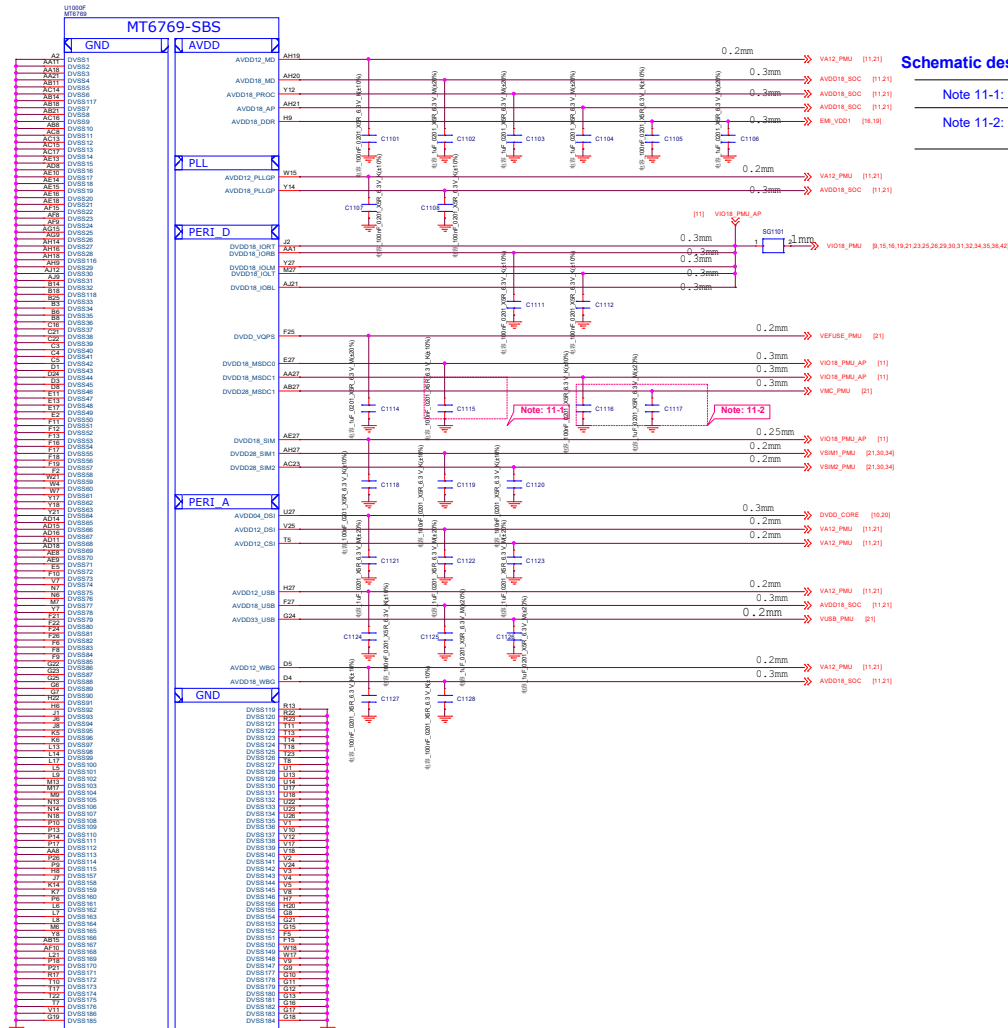
# Ship Mode Det





# Schematic design notice of "10\_BB\_POWER\_PDN" page:

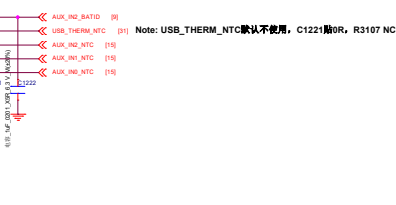
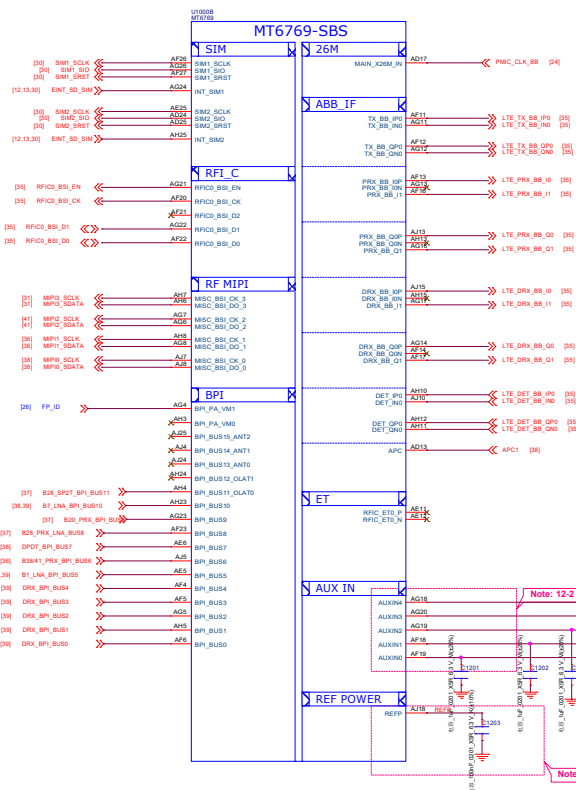
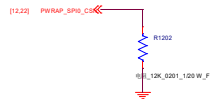
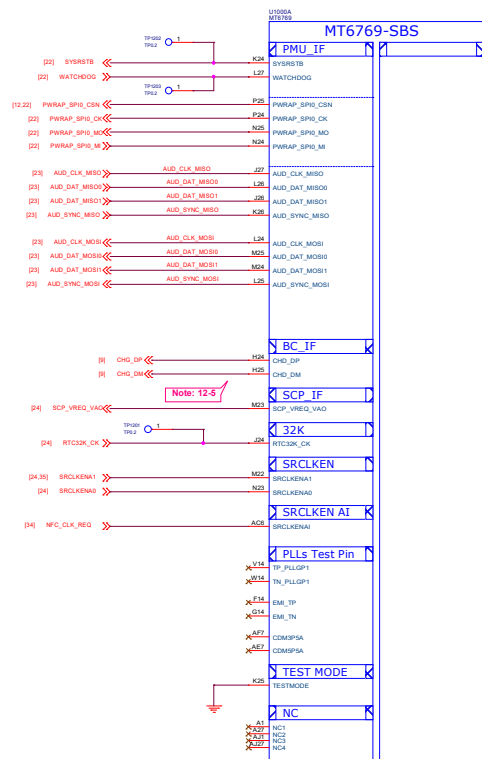
- Note 10-1: Differential pair of DVDD\_GPU remote sense must be close to BB's ball.  
Remote sense trace with GND shielding to PMIC (Differential)
- Note 10-2: Differential pair of DVDD\_PROC remote sense must be close to BB's ball.  
Remote sense trace with GND shielding to PMIC (Differential)
- Note 10-3: Differential pair of DVDD\_MODEM remote sense must be close to BB's ball.  
Remote sense trace with GND shielding to PMIC (Differential)
- Note 10-4: Differential pair of DVDD\_CORE remote sense must be close to BB's ball.  
Remote sense trace with GND shielding to PMIC (Differential)



Schematic design notice of "11\_BB\_POWER\_IO" page:

Note 11-1: C1115 closed DVDD18\_MSDC0 150mil

Note 11-2: C1116 closed DVDD18\_MSDC1 150mil  
C1117 closed DVDD28\_MSDC1 150mil



## Schematic design notice of "12\_BB\_1" page:

Note 12-1: PWRAP\_SPI0\_CS\_N and "AUD\_DAT\_MOSI0" are bootstrap pins to select which interface will be the JTAG pin out

| PWRAP_SPI0_CS_N  | AUD_DAT_MOSI0    | JTAG Function            |
|------------------|------------------|--------------------------|
| default=PU       | default=PD       | AP_JTAG MD_JTAG          |
| HI               | LO               | N/A                      |
| HI               | HI (by ext. PDI) | SPI0 + EINT8 SPI2 + SPI3 |
| LO (by ext. PDI) | LO               | SPI0 + EINT8 N/A         |
| LO               | HI (by ext. PDI) | N/A                      |

Note 12-2: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

Note 12-3: The de-coupling cap. for REFP (AJ18 ball) have to be placed as close to BB as possible.

Note 12-4: HW pin for DDR type feature in bootstrap (refer to BB HW design Notice)

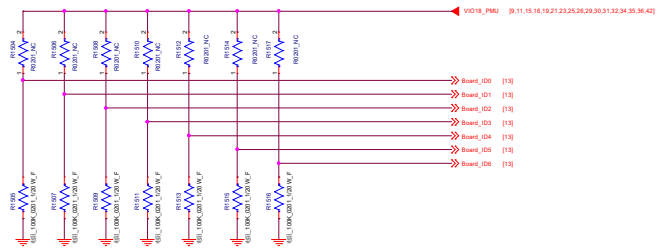
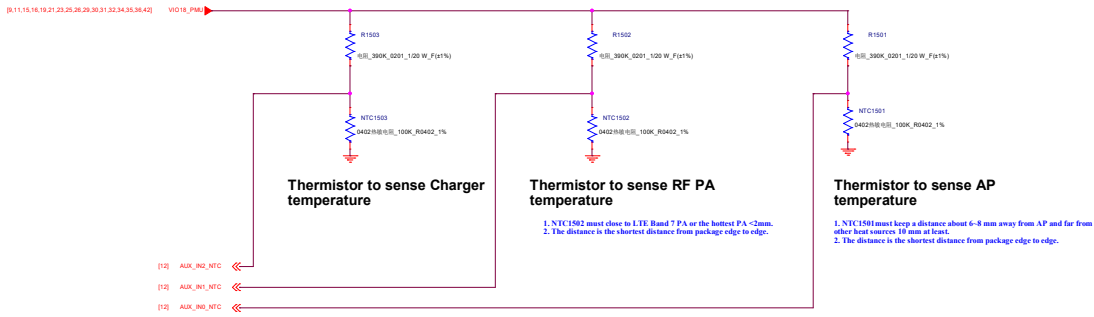
| AUD_SYNC_MISO     | AUD_CLK_MISO      | CAM_PDN3     | PMIC 658 voltage | DDR Type  |
|-------------------|-------------------|--------------|------------------|-----------|
| default=PD        | default=PD        | default=PD   | VDRAM1 / VDRAM2  | DDR       |
| No ext. pull      | No ext. pull      | No ext. pull | 1.125v / 0.6v    | LP4X eMCP |
| No ext. pull      | 12K pull to VIO18 | No ext. pull | OFF / 1.8v       | Reserved  |
| 12K pull to VIO18 | No ext. pull      | No ext. pull | 1.225v / 0.7v    | LP3 eMCP  |
| 12K pull to VIO18 | 12K pull to VIO18 | No ext. pull | 1.125v / 1.8v    | Reserved  |

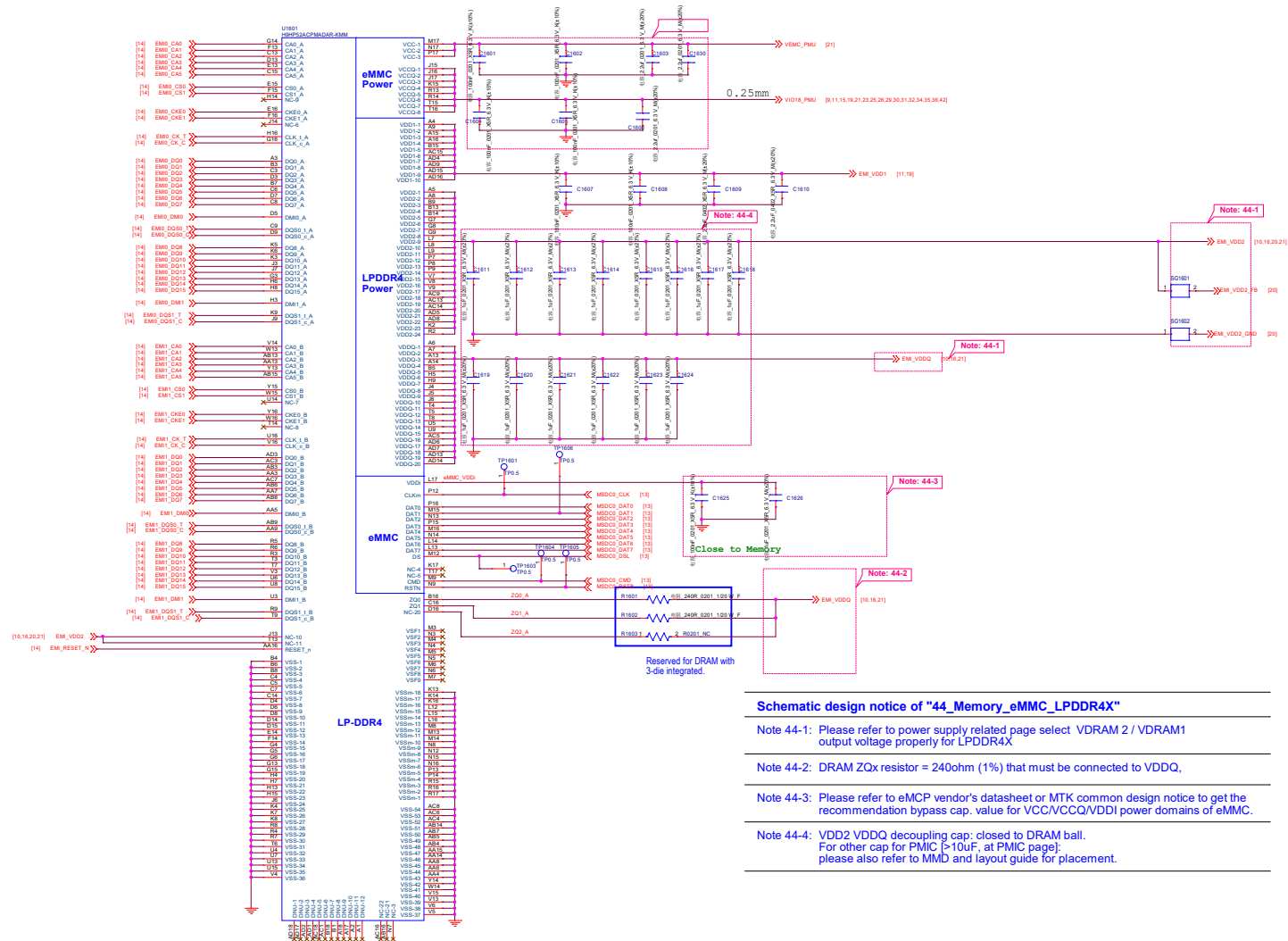
Note 12-5: Charger must have D+/D- pin for charger type USB detection. Charger must have at least 500mA USB current for All charger type.





Note 14-1: R1401 please select 60.4 ohm (1%) resistor





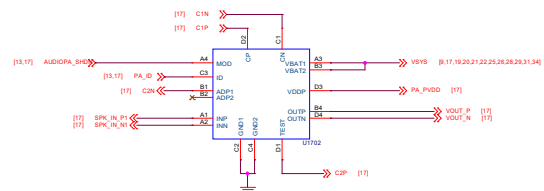
# Schematic design notice of "44\_Memory\_eMMC\_LPDDR4X"

Note 44-1: Please refer to power supply related page select VDRAM 2 / VDRAM1 output voltage properly for LPDDR4X

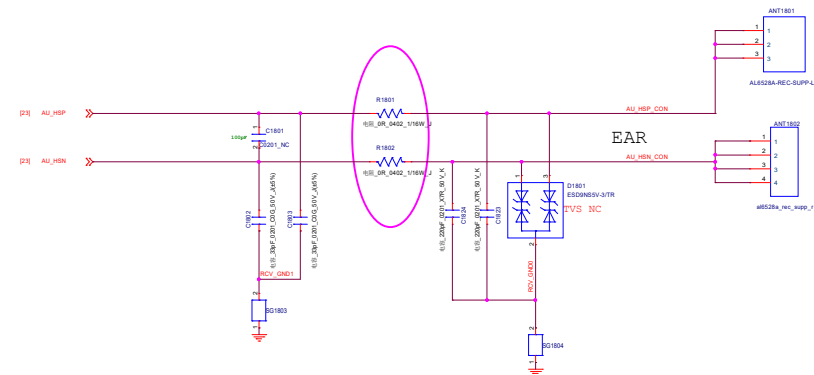
Note 44-2: DRAM ZQx resistor = 240ohm (1%) that must be connected to VDDQ,

Note 44-3: Please refer to eMCP vendor's datasheet or MTK common design notice to get the recommendation bypass cap. value for VCC/VCCQ/VDDI power domains of eMMC.

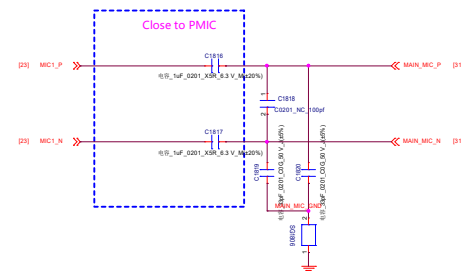
Note 44-4: VDD2 VDDQ decoupling cap: closed to DRAM ball.  
For other cap for PMIC (~10uF, at PMIC page);  
please also refer to MMD and layout guide for placement.



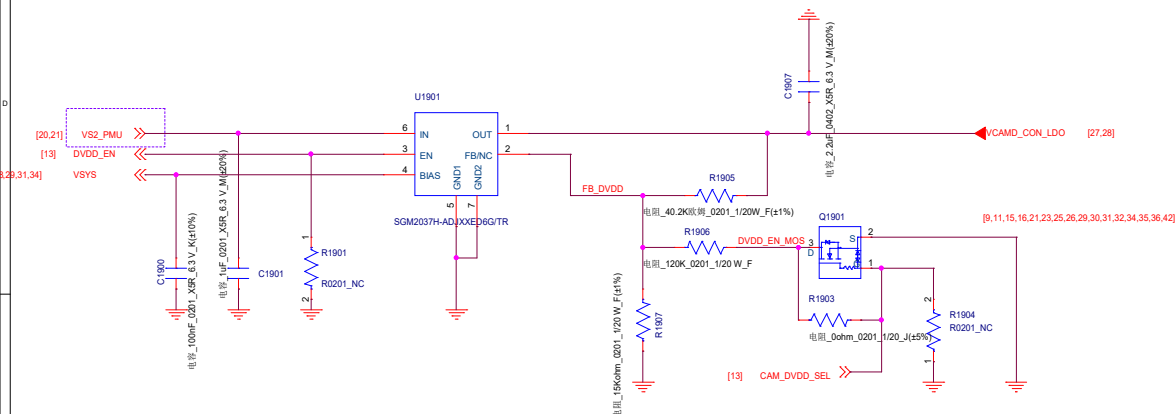
## RECIVER



## MAIN MIC KB

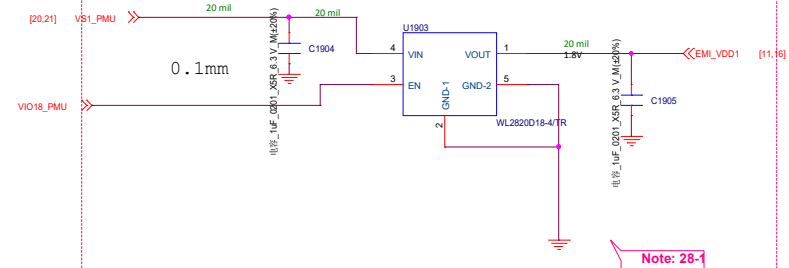


### LDO forUM 5M(VCAMD\_EXT1) 1.2V

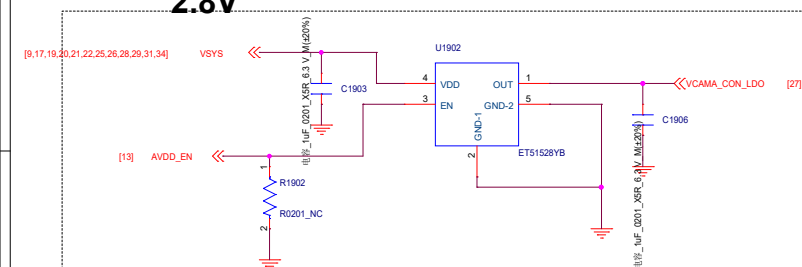


|       |       |       |       |       |       |      |            |
|-------|-------|-------|-------|-------|-------|------|------------|
| R1905 | R1906 | R1907 | Q1901 | R1903 | R1904 | 电压   | CAM_DVDD_B |
| ON    | ON    | ON    | NC    | ON    | NC    | 1.1V | Float      |
| ON    | ON    | ON    | NC    | ON    | NC    | 1.2V | L          |
|       |       |       |       |       |       |      |            |
|       |       |       |       |       |       |      |            |

### Ext. EMI\_VDD1 for LPDDR4X VDD1

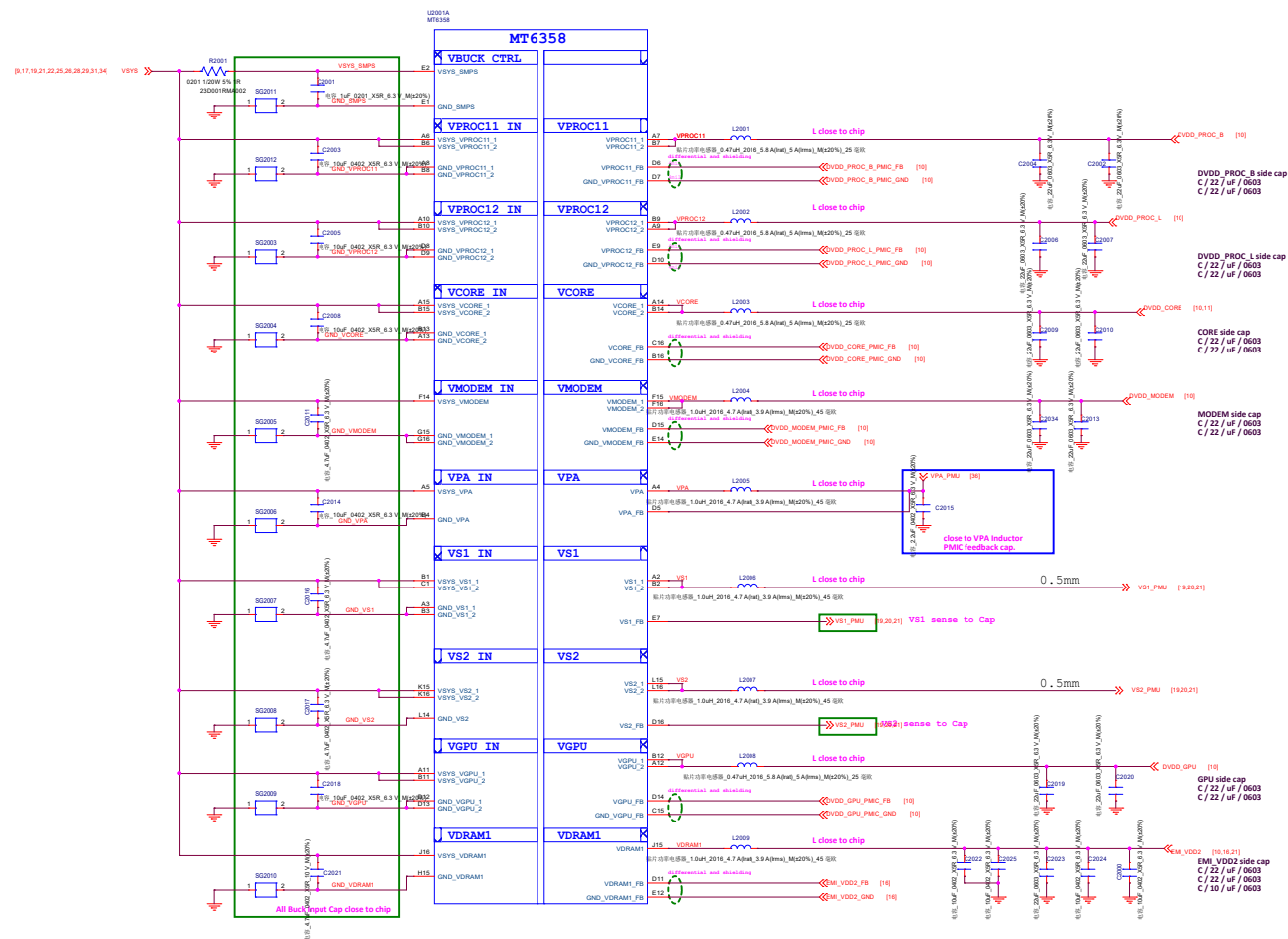


## LDO for Dual Camera (VCAMA\_EXT) 2.8V



**Schematic design notice of "28\_POWER\_ThirdParty\_Powers" page:**

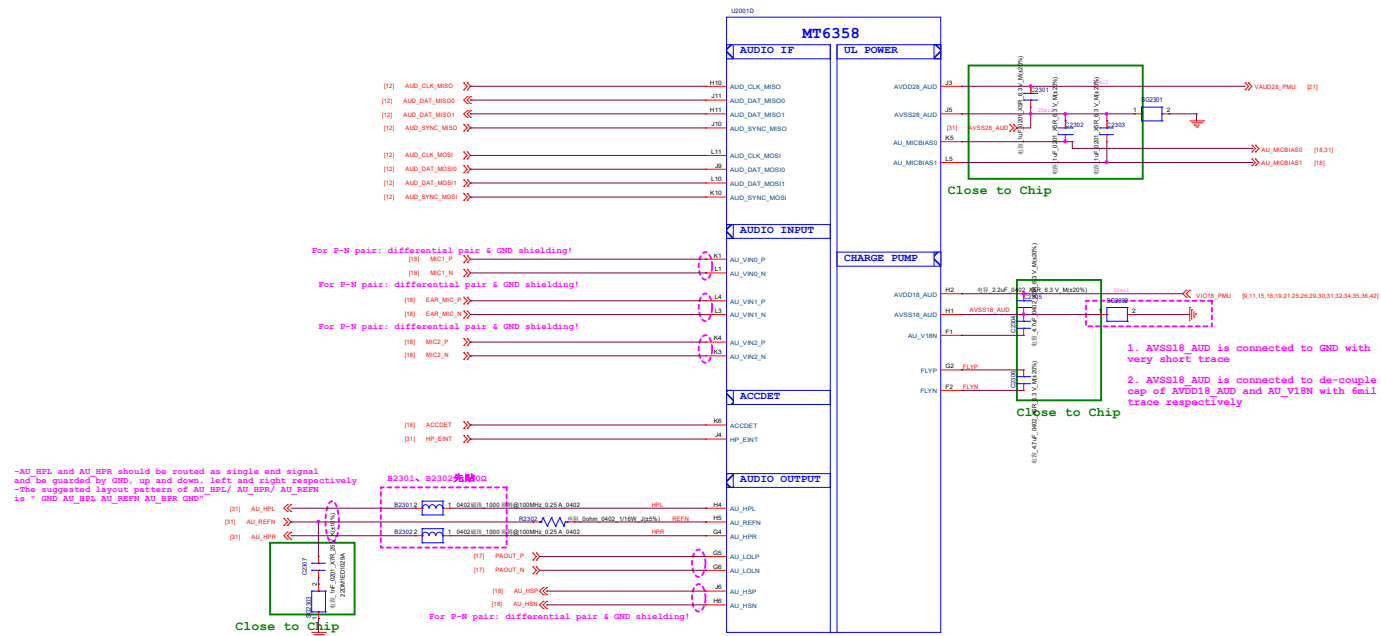
Note 28-1: Refer to BB design notice for LP4X VDD1 LDO selection guide





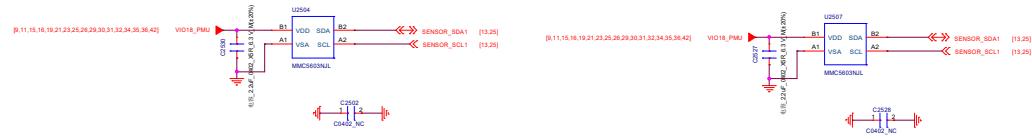




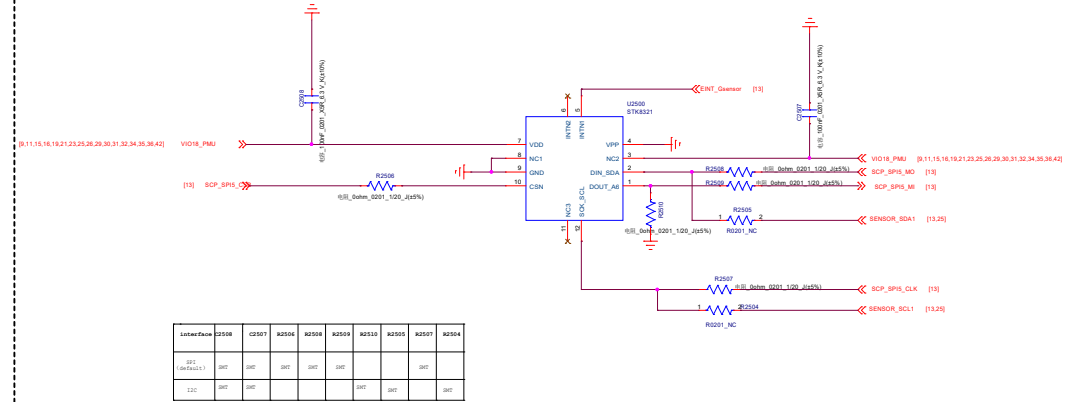




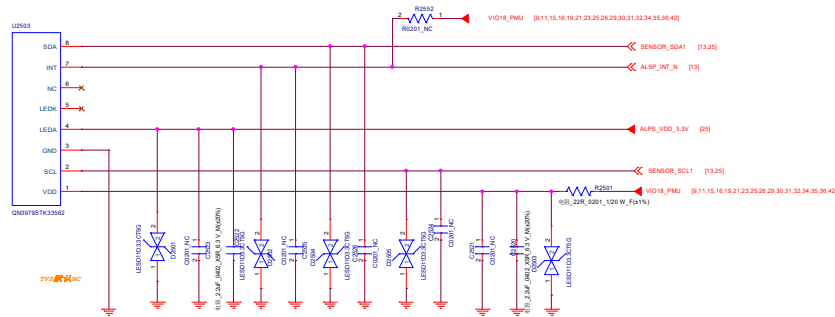
# eCOMPASS



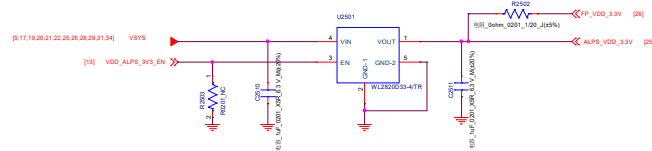
# G SENSOR



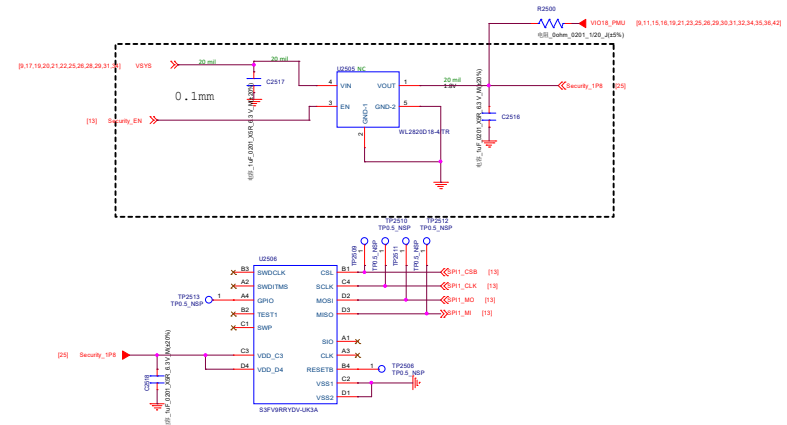
# ALPS Sensor



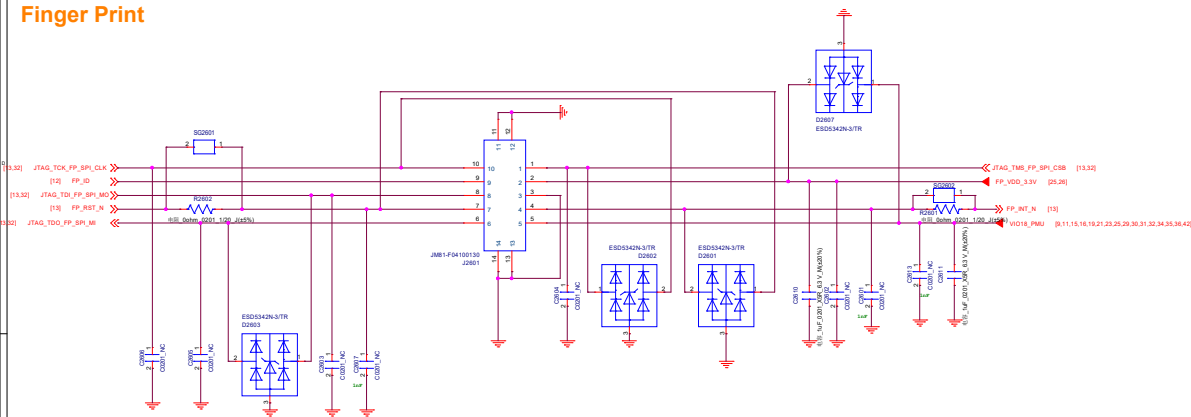
# LDO For ALP



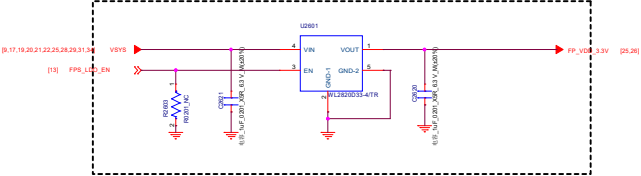
# Security IC



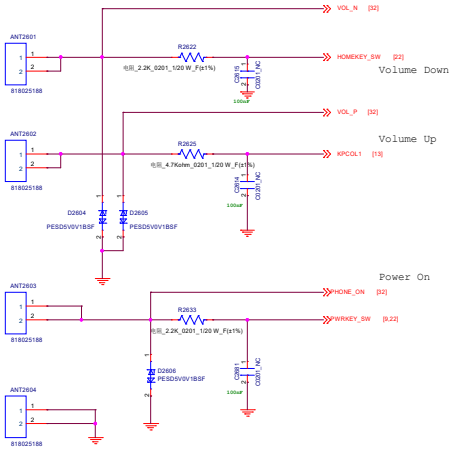
Finger Print



LDO for FPS

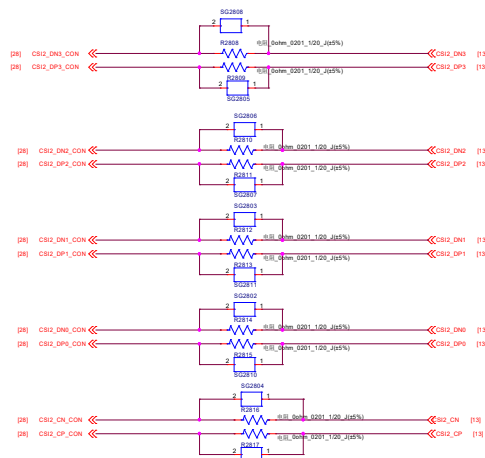
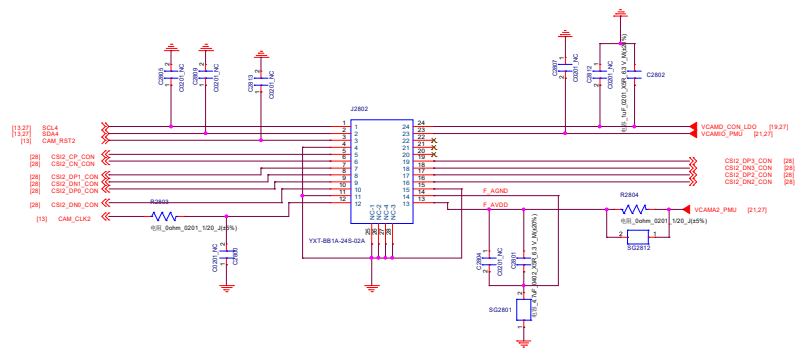


KEY pad

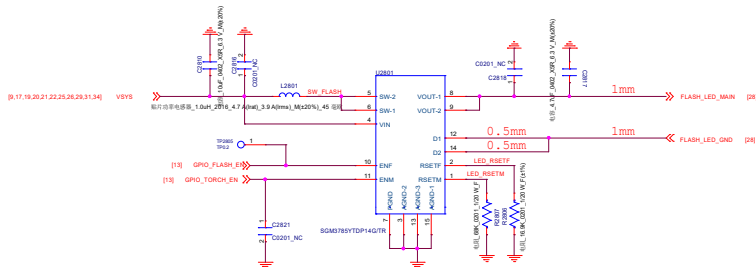


[illegible][illegible]

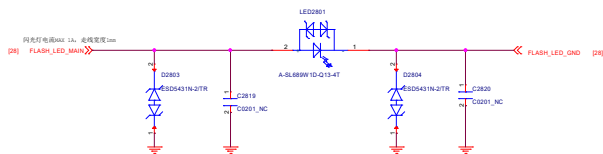
Front 13M Cam



Flash Driver

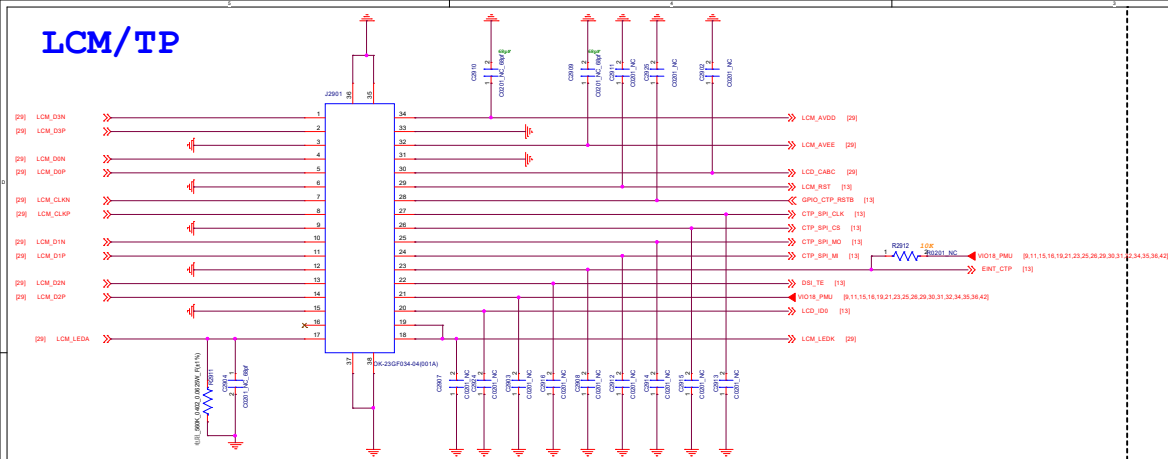


Flash LED

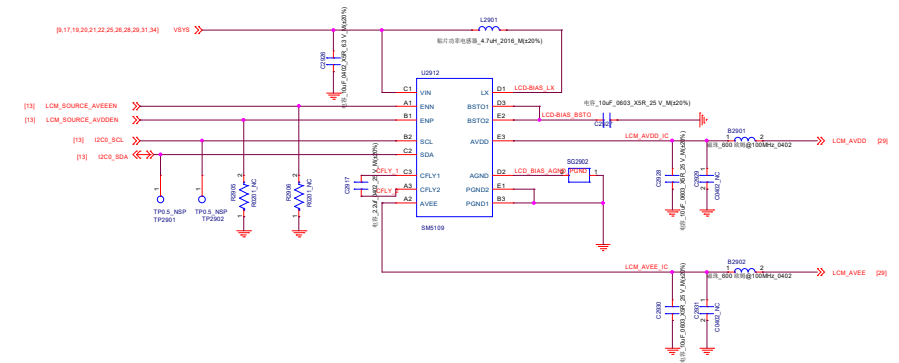




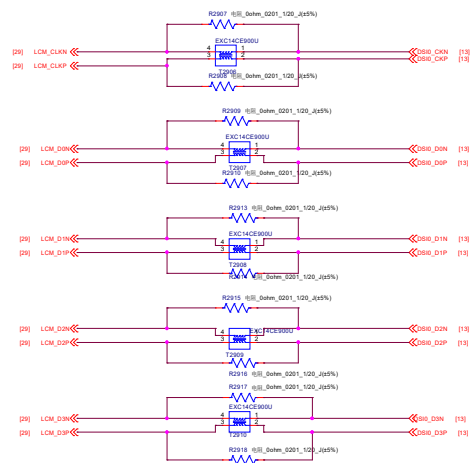
LCM/TP



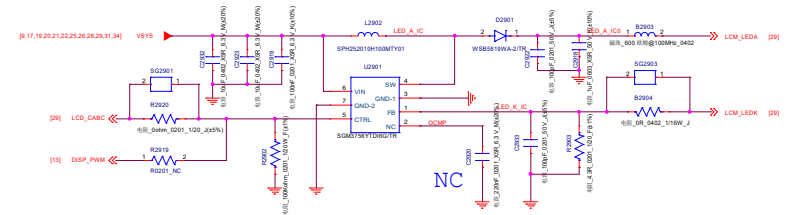
## LCM Bias



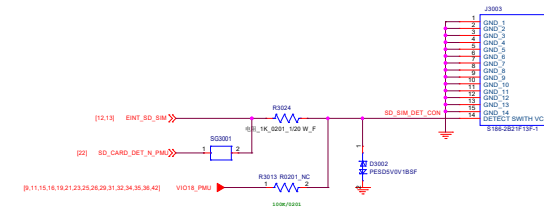
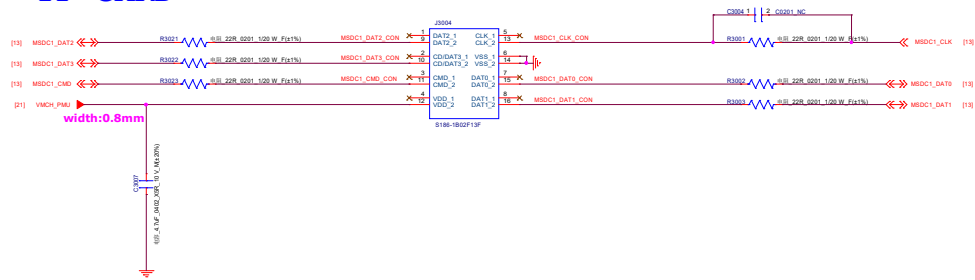
## LCD MIPI Common Mode Filter



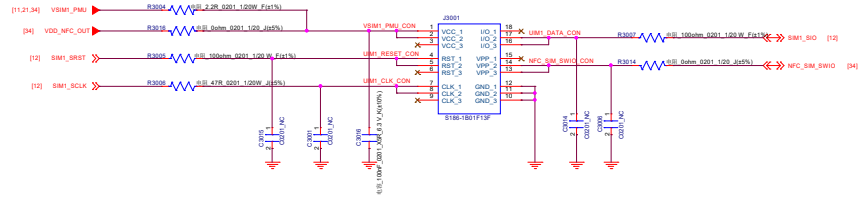
## LCM Backlight Driver



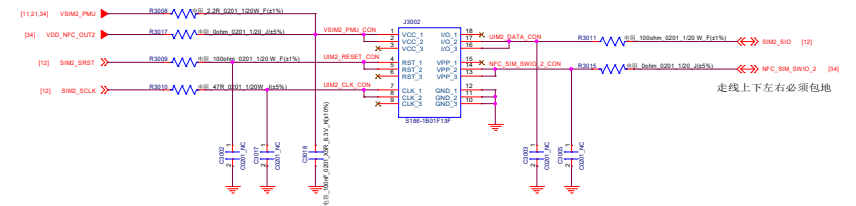
**TF CARD**



## NANO SIM1

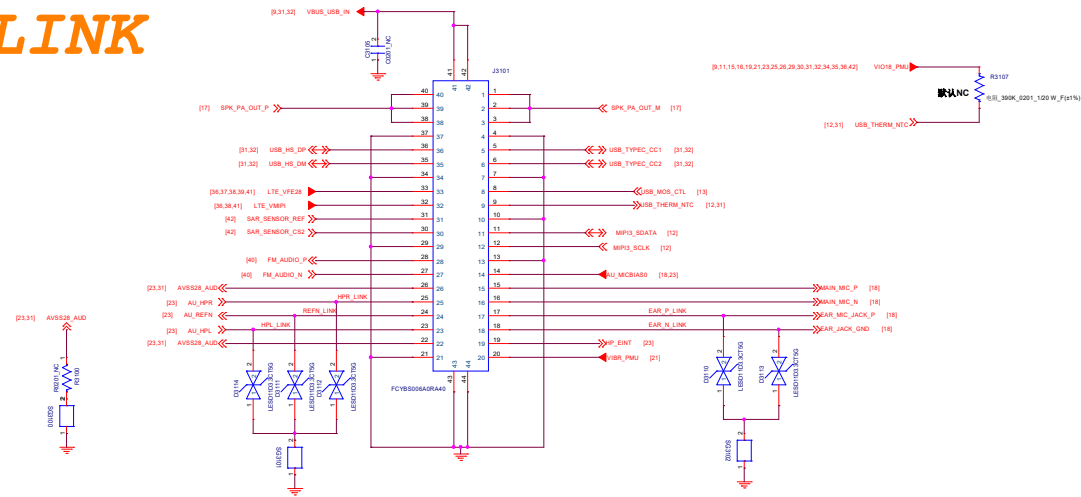


NANO SIM2

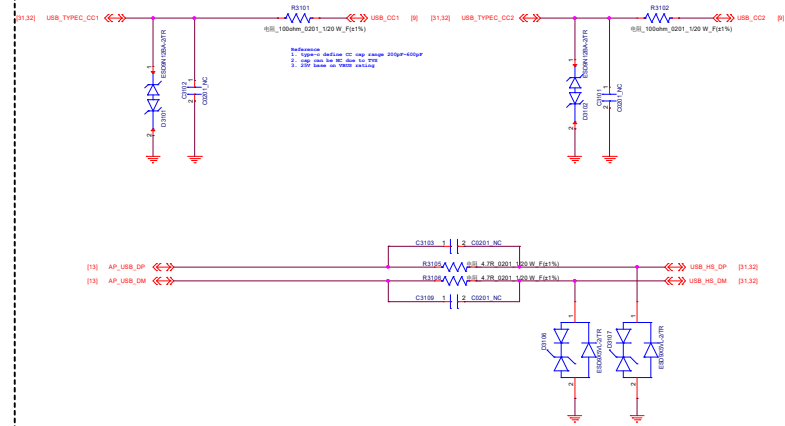


走线上下左右必须包地

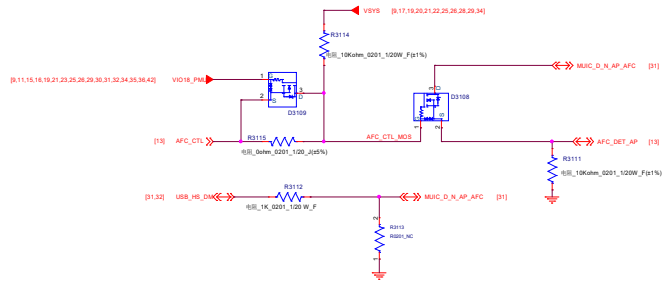
# LINK



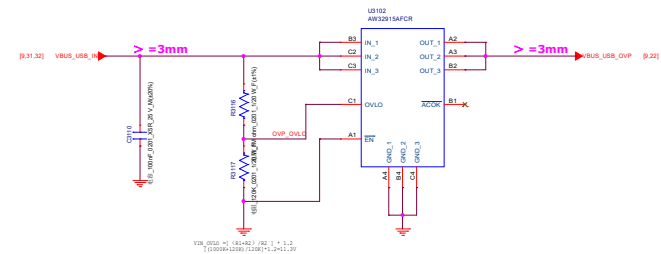
# CC&DP / DM



# AFC



# OVP

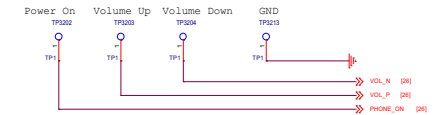


VBAT

**MARK**

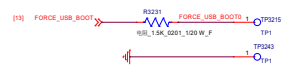


## KEYPAD

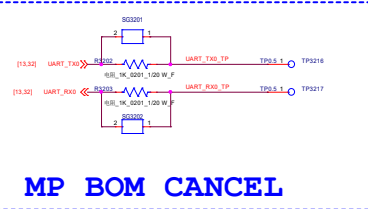


## JTAG Connector

# FORCE DL

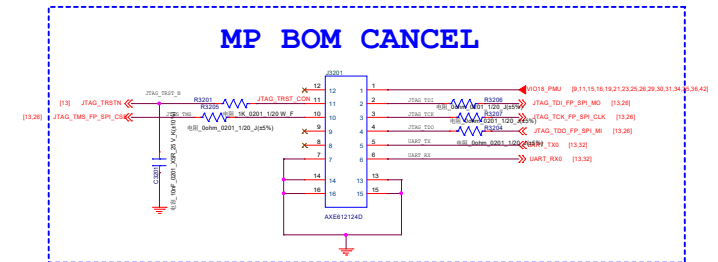
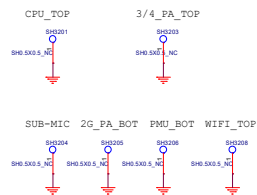


## UART

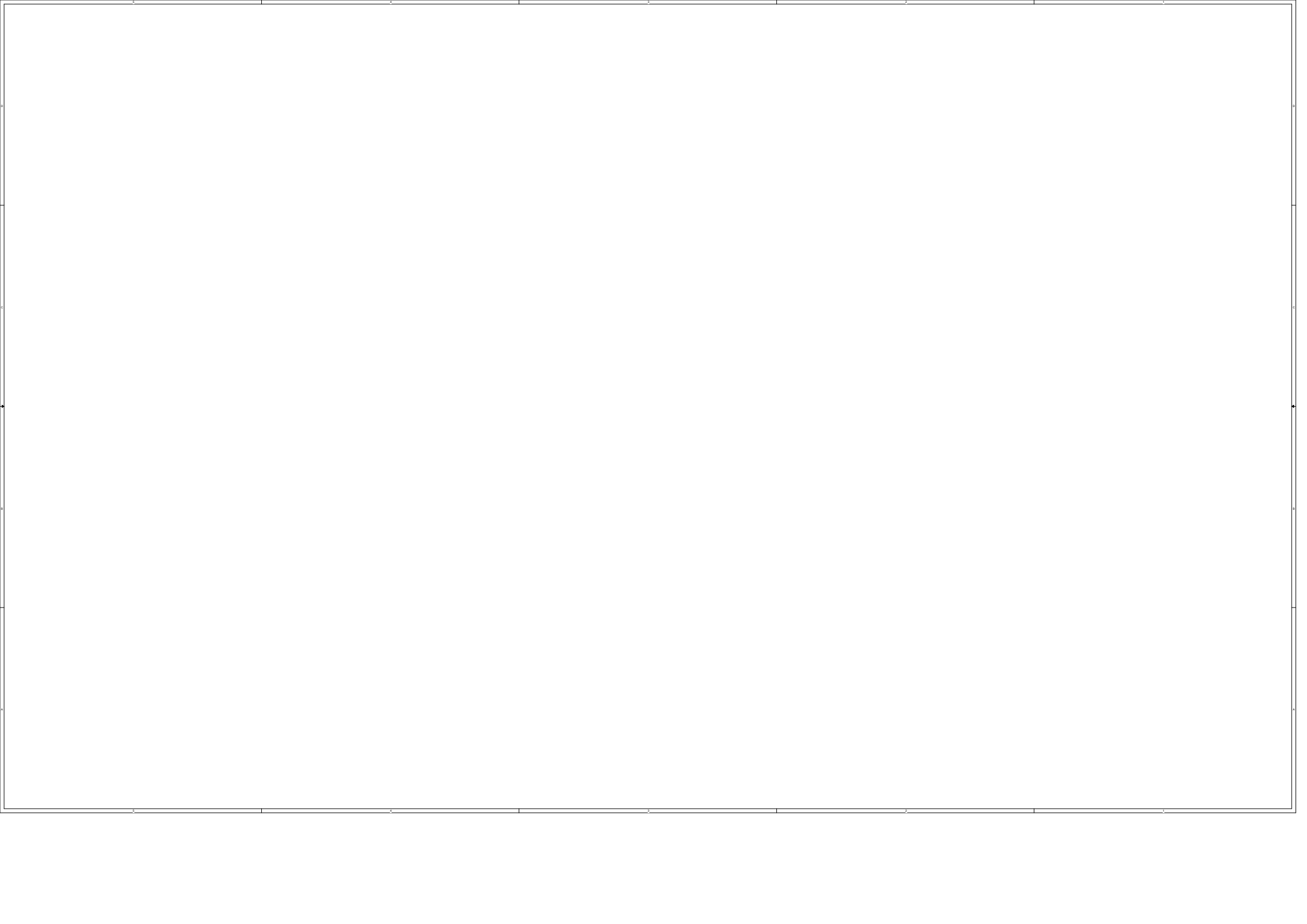


[13] AP\_GOOD TP0.5 1 TP321

## Shielding GND

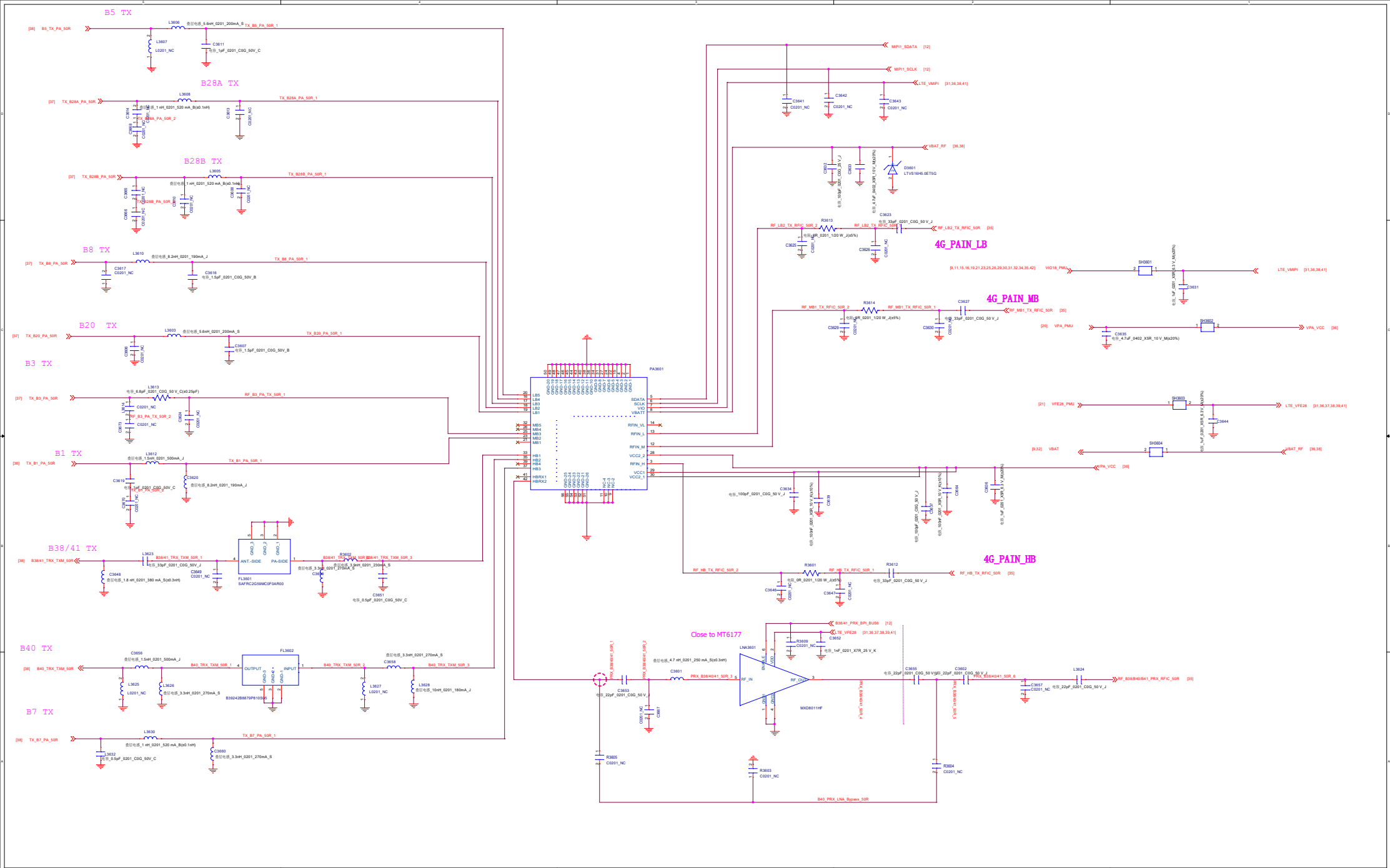


J1201连接座换成HQ11910599000 OK-14GM012-04(001A),



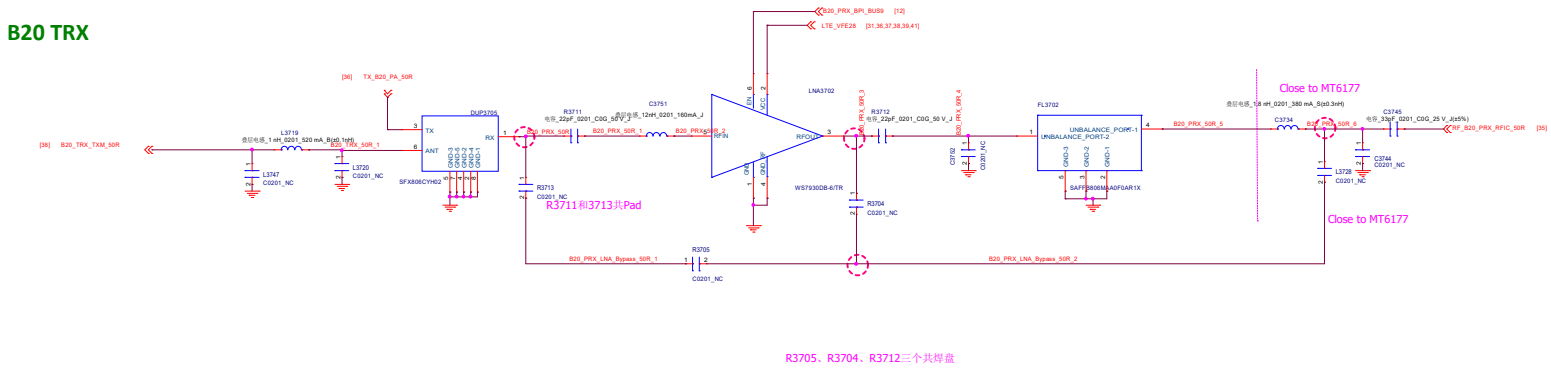




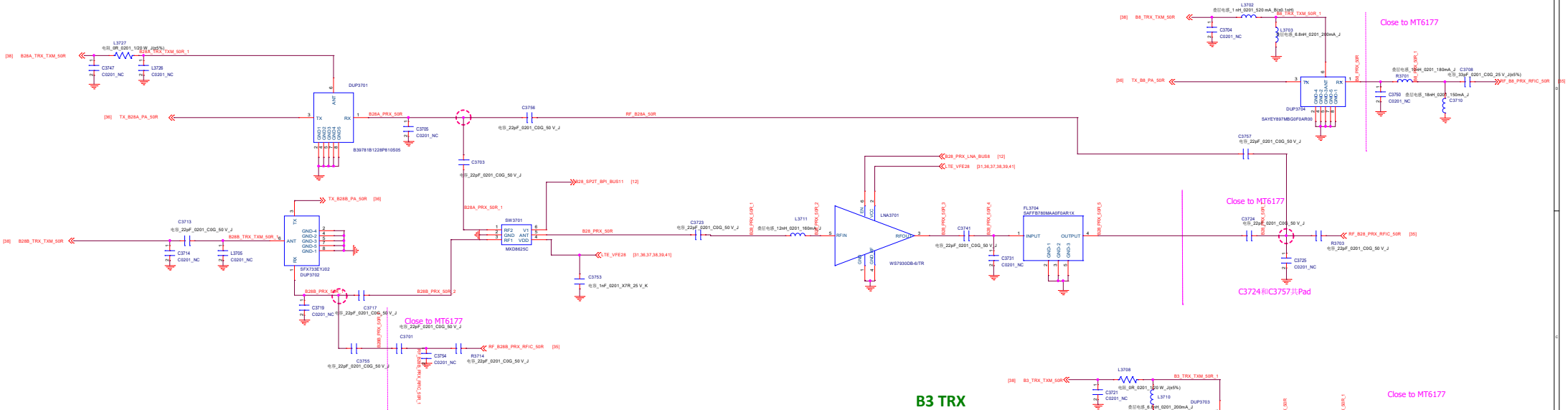




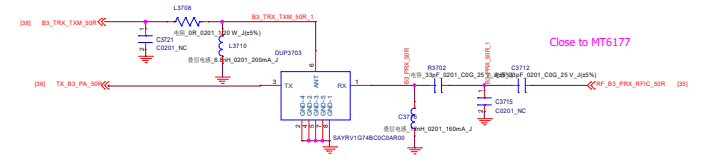
## B20 TRX



**B8 TRX**

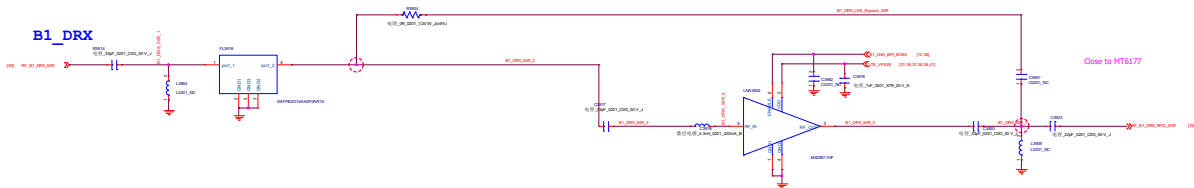


**B3 TRX**

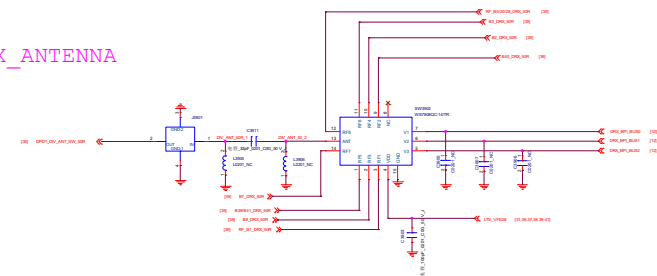




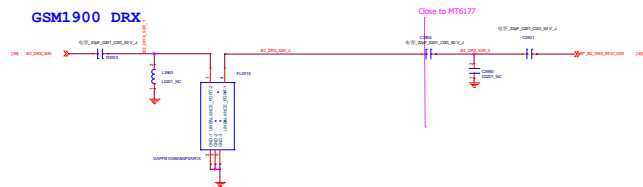
B1\_DRX



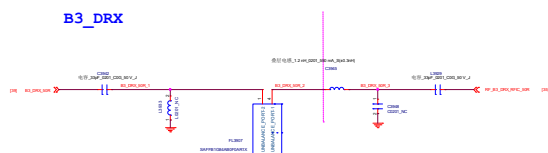
DRX\_ANTENNA



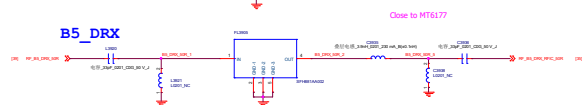
GSM1900\_DRX



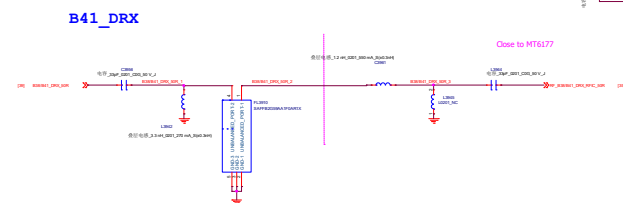
B3\_DRX



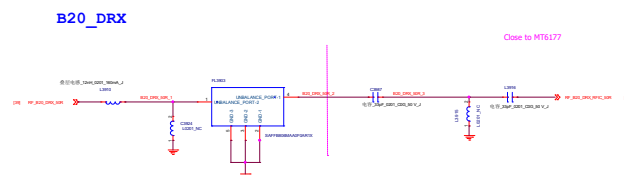
B5\_DRX



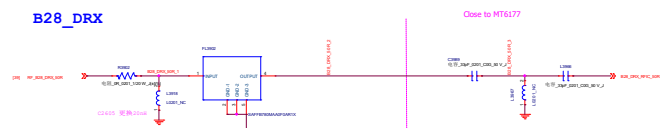
B41\_DRX



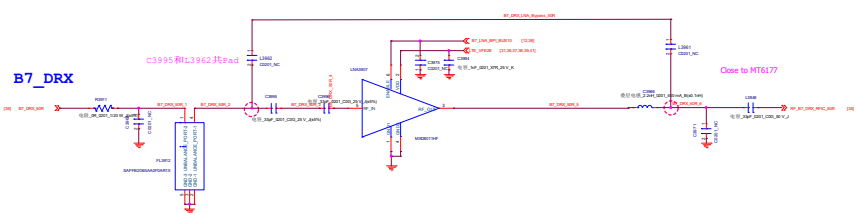
B20\_DRX



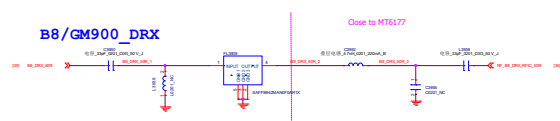
B28\_DRX



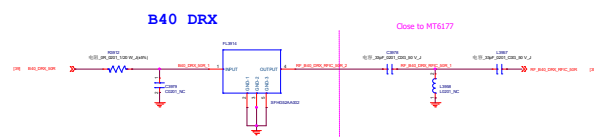
B7\_DRX

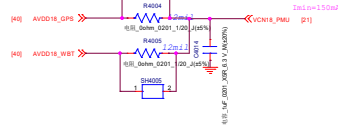


B8/GM900\_DRX



B40\_DRX

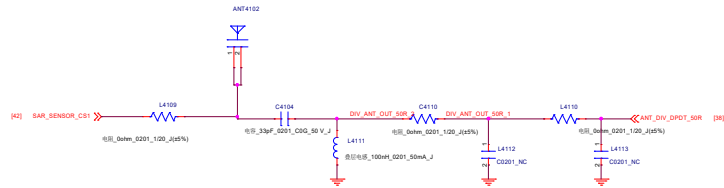




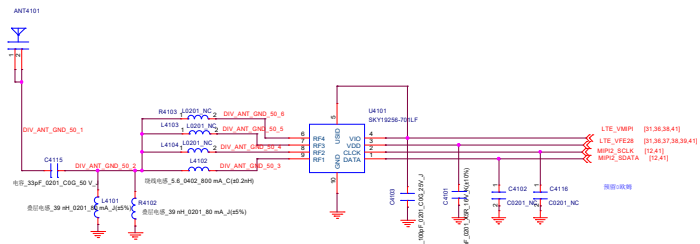
**Note 51-4:** Pin 36 (AVDD28\_FM) must be connected to VCN28 even if FM not support

**Note 51-4:** Pin 36 (AVDD28\_FM) must be connected to VCN28 even if FM not support

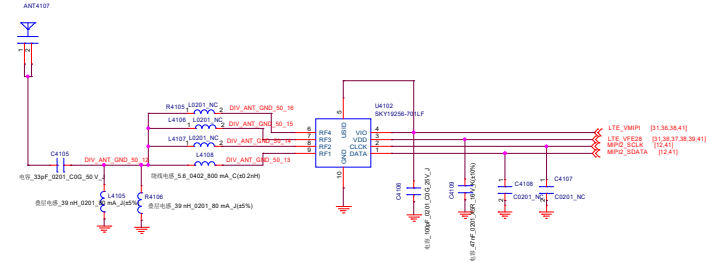
DIV ANT



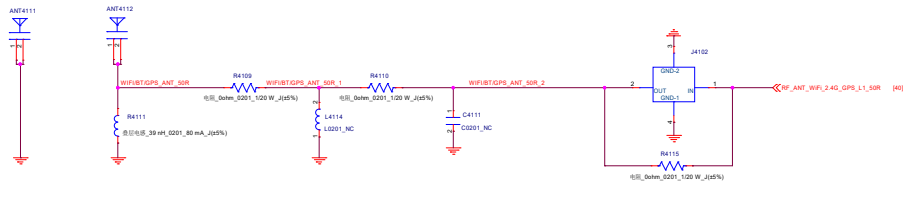
DIV GND



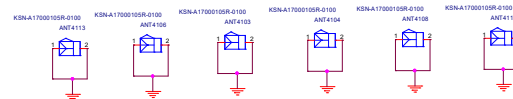
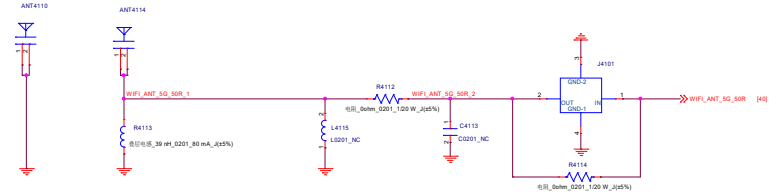
DIV GND

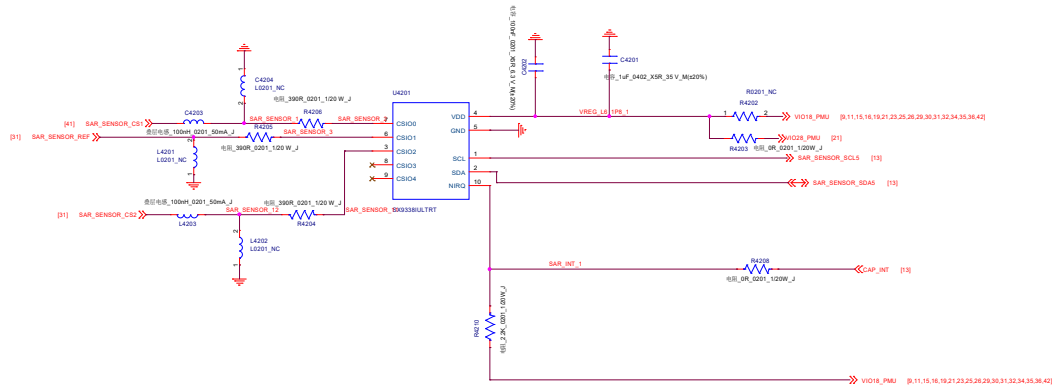


2.4G\_GPS ANT

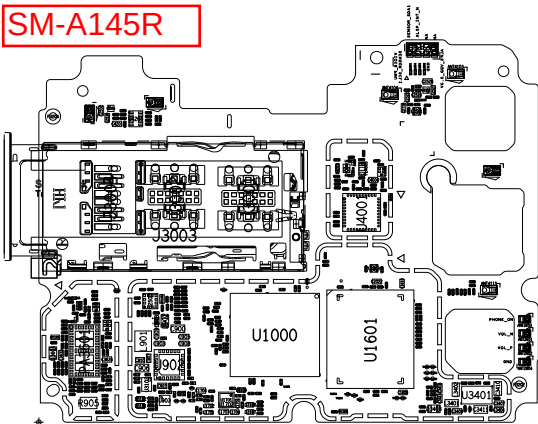


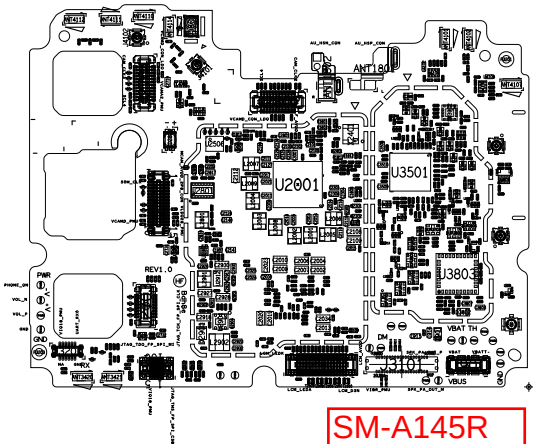
## Wifi 5G ANT





SM-A145R







# SM-A145P/R

